



## CASE STUDY:

# SOUTHEASTERN PENNSYLVANIA TRANSPORTATION AUTHORITY (SEPTA)

Southeastern Pennsylvania Transportation Authority (SEPTA) utilizes non-intrusive underground mapping technology to complete the coordination of underground utilities for a railroad station parking garage in suburban Pennsylvania.



LOCATION: CONSHOHOCKEN, PA, USA



DATE: APRIL 2023



TERRAIN: SUBURBAN



PROJECT PHASE: PRE-CONSTRUCTION



USE CASE: PARK-AND-RIDE  
CONSTRUCTION

## TRANSPORTATION AUTHORITY NEEDED TO IDENTIFY AND RELOCATE WATER MAIN TO CONTINUE CONSTRUCTION OF COMMUTER RAIL PROJECT

- SEPTA needed to identify the location of an existing underground water main at the future Conshohocken Station parking garage along its Manayunk/Norristown commuter rail line to coordinate the foundation supports for the parking garage.
- The owner of the main, Aqua Pennsylvania, Inc.'s water service utility maintenance team recommended relocation of the main from under the parking garage footprint. Therefore, the main needed to be accurately surveyed, and a new water main location designed to relocate it from under the parking garage.
- Ground conditions had changed drastically since the line's original installation in 1955 and Aqua did not have accurate as-built survey information to identify its location.
- The exact depth of the line was estimated to be 25-30 feet but also unknown because the previous property owner had filled the site with over 15 feet of soil and construction debris such as brick and asphalt, rendering conventional locating methods ineffective.

## EXODIGO PROVIDED A COMPLETE UNDERGROUND MAP TO STREAMLINE FINAL DESIGN DELIVERY FOR THE PROJECT

- SEPTA engaged Exodigo to locate the water main and provide a complete map of underground assets in the project area.
- Exodigo scanned every square foot of the general area where the main was estimated to be located with multiple sensors, camera images, and centimeter-precision RTK GPS to collect over 500 GB of data per acre.
- The data was then analyzed and processed with AI and turned into a visual underground network model that allowed SEPTA to "see" the underground in its entirety, including the location of the water main.

## AI-POWERED INPUTS TO MAP THE UNDERGROUND ACCURATELY



### Best Available Visuals

3D geolocated high-resolution image capture + satellite images and web sources



### Records + As-Built

AI-driven digitalization of analog formats + automated search and collection of GIS layers



### Platform

Systematic scanning + data collection across the project site with our cart platform.



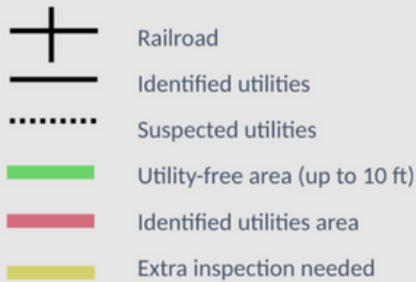
### Sensors

Signal processing across magnetic gradiometer, TDEMI and FDEM, multi-channel GPR, and electromagnetic sensors

Exodigo uses AI to transform thousands of data points into a visual map of the underground, showing utility-free "Green Zones," possible utility "Yellow Zones," and identified utilities, overlaid on satellite imagery of the site.

ADDITIONAL SUSPECTED  
UTILITIES MARKED

WATER MAIN CORRECTLY  
IDENTIFIED AT 25 FT DEPTH  
RUNNING UNDER TRACKS



## WATER MAIN CORRECTLY LOCATED 30 FEET EAST OF ASSUMED LOCATION

The drastic changes in site conditions since the water main was buried in the 1950s made it difficult - if not impossible - to use the geographic references in the records to dig in the accurate location.

In April 2023, after failed attempts to positively identify the water main via a traditional utility locating approach, Aqua's team dug at the water main's last known location per their records. However, after 35 feet of excavation, the water main was not identified at this location.

Simultaneously, SEPTA engaged Exodigo to attempt to locate the water main. Exodigo conducted a multi-sensor scan and detected the line ~30 feet east of Aqua's assumed location.

Aqua excavated at the Exodigo-identified location and encountered the water main at a depth of 25 feet - precisely where Exodigo advised it would be.

As a result, Aqua has now located both the north and south ends of the pipe. Without Exodigo's insights, the crew would have had to trench the entire site, which would have taken weeks. Instead, they only used an excavator for four days, and are now completing the design of the water main relocation.

### BY THE NUMBERS

# ~3 WEEKS SAVED

*By using Exodigo's insights, SEPTA avoided weeks of heavy trenching.*

***"Based on these results, and additional work that Exodigo has performed with other transit agencies, we believe there may be opportunities to expedite and ensure more accurate existing underground utility survey information for project delivery across multiple divisions at SEPTA by leveraging Exodigo's innovative technology and approach to subsurface utility location mapping."***

Robert Tangi, Project Manager, SEPTA